

Config			Rise time		Settling		Overshoot		Steady-state error	
α	k_p	k_i	$T_r^\uparrow$ (f)	$T_r^\downarrow$ (f)	$T_s^\uparrow$ (f)	$T_s^\downarrow$ (f)	OS $^\uparrow$ (%)	OS $^\downarrow$ (%)	SSE step (ms)	SSE saw. (ms)
0.83	2.0	0.33	17	10	15	8	14	39	0.414	0.591
0.85	2.0	0.33	17	8	15	5	14	43	0.401	0.569
0.93	2.0	0.33	13	6	29	26	53	83	0.454	0.970
0.93	2.0	2.0	11	7	26	22	47	62	0.369	0.893
0.93	4.0	0.80	13	7	27	23	42	60	0.349	0.930

under-smoothed

chosen

over-smoothed

high k_i

high k_p

Metrics: T_r = rise time (90% of step, causal MA-20), T_s = settling (± 1.5 ms, causal MA-20), OS = overshoot, SSE = mean |realized−target| (centered MA-30). $\uparrow$ = step-up budget, $\downarrow$ = step-down. α = chosen default ($\alpha=0.85$, $k_p=2.0$, $k_i=0.33$).